

Neuroinformatics - Project 1

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1 Introduction and background

High-resolution Magnetic Resonance Imaging (MRI) is essential for accurate clinical diagnosis and neuroinformatics research. However, acquiring high-resolution 3D volumetric data is time-consuming, expensive, and increases the likelihood of motion artifacts due to prolonged scan times. While accelerating the acquisition process by taking low-resolution scans mitigates these issues, it significantly compromises diagnostic image quality and structural detail.

Historically, tackling this problem relied on traditional interpolation methods (e.g., bicubic or B-spline interpolation) which fail to recover high-frequency details and often result in blurry images [7]. Recent advancements in deep learning, specifically Generative Adversarial Networks (GANs) and foundation models, have provided new avenues to solve this [1, 5].

By framing this as a 3D super-resolution (SR) problem, we aim to algorithmically upscale low-resolution (LR) MRI volumes to high-resolution (HR) outputs. We hypothesize that by adapting 2D super-resolution architectures into the 3D domain (such as 3D ESRGAN [10]), we can capture inter-slice spatial dependencies that 2D models miss. The current availability of high-compute clusters (utilized via SLURM) and advanced volumetric loss metrics make now the ideal time to solve this issue. Our approach aims to overcome previous blurring obstacles by incorporating a dedicated perceptual/metric model to ensure the generated voxels are not just mathematically accurate but visually and diagnostically realistic.

2 Datasets

2.1 Preprocessing

Original Data:

The divergence between low-field (LF) and high-field (HF) MRIs is well studied and lies in the fundamental scaling of the Signal-to-Noise-Ratio (SNR) and its subsequent impacts on spatial resolution. From literature, and through observing our dataset, we can appreciate the following main characteristics:

1. Spatial Resolution:

- **Morphological Blurring and Signal Attenuation:** LF acquisitions (typically $< 0.5T$) show pronounced blurring and loss of fine structural detail. This is largely a consequence of the reduced SNR—which scales approximately the strength of the magnetic field with an additional dependence on the dominance of body vs. coil noise [4]. Due to this, to maintain adequate signal, LF systems necessitate larger voxels, which inherently limits spatial resolution. This in turn, reduces the sharp definition observed in the HF "ground truth" references.
- **Contrast Compression and Tissue Differentiation:** LF imaging frequently suffers from a compressed dynamic range. While HF imaging provides distinct intensity separation between gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF), the overlapping T_1 and T_2 relaxation times at lower field strengths lead to "signal bleeding." This degradation of the Contrast-to-Noise Ratio (CNR) significantly complicates downstream tasks like automated tissue segmentation [8].

2. Stochastic Noise:

- **Rician Distribution Dynamics:** MRI magnitude data follows a Rician distribution [3]. In the low-signal regimes characteristic of LF imaging, this distribution becomes highly skewed. This introduces a non-zero mean bias in the background that can artificially inflate intensity measurements and hinder the performance of standard denoising algorithms.
- **Susceptibility and Geometric Distortion:** While LF systems are inherently less sensitive to certain macroscopic susceptibility artifacts (such as those arising from metallic implants), they

remain vulnerable to geometric distortions caused by inhomogeneities in the static magnetic field (B_0 inhomogeneities) [6].

3. **Spectral Incompleteness and Class Imbalance** Although touched upon above, the spectral incompleteness and class imbalance is a large reason why this problem is so difficult to solve, as sparsity and class overlap make linear interpolation impossible between both sets of MRIs.

- **Frequency Domain Sparsity:** In the k-space domain, LF images typically lack high-frequency data. These missing components correspond to the "edges" and fine textures of the anatomy. At low field strengths, these features are often buried beneath the noise floor, making them irrecoverable through simple linear interpolation.
- **Statistical Distribution and Class Overlap:** Analysis of intensity histograms reveals that LF distributions often tend toward uni-modality. In contrast, HF histograms exhibit distinct peaks corresponding to specific tissue classes [2]. The "squeezing" of these classes in LF data creates a significant data imbalance; the dominant background and noise signals overwhelm the biologically relevant signals.

Preprocessing steps (preprocessing.py).

While viewing slices of the original volumes, along with their corresponding intensity distributions, we observed many of the above factors mentioned in the literature. In an attempt to mitigate the above factors and prepare the MRI volumes for training, several preprocessing steps were applied to the raw NIfTI data:

- **Intensity Normalization:** Voxel intensities were normalized to the standard range of $[\epsilon, 1]$ (where $\epsilon = 1e - 8$ to prevent divisions by zero) to stabilize training.
- **Patch Extraction:** Due to the memory constraints of processing full 3D volumes, we decided that the most optimal approach was to split the volumes into overlapping 3D patches (e.g., $64 \times 64 \times 64$ voxels) that were extracted from the scans.
- **Smart Tissue Sampling (Foreground Masking):** Utilizing our patches we could avoid feeding the foreground pixels to the model. Upon loading our patches, we would guaranty that at least 20% of the extracted patch contained actual tissue or the cranium.
- **Spatial Alignment:** Due to the different dimensions of the scans - $1.6 \times 1.6 \times 5.0mm$ vs. $1.0 \times 1.0 \times 1.0mm$ for the 3T scan - we resampled from both volumes to dynamically resize and align the LF and HF volumes to assure that voxel locations were anatomically relevant.

These steps mainly looked to attack the differences in spatial resolution, stochastic noise and class imbalances in order to create more uniform and understandable dataset which we expected would be better processed and "learned" by the model. The applied steps, with the exception of the tissue sampling and patch extraction — which are limited to model training —, can be observed in figure 1 in which we show 4 separate slices of training sample 11 to show the alignment, and normalization steps that have already been applied. If the alignment step is not applied previously, it is very difficult to visualize similar region slices as the different voxel sizes lead to different tensor sizes.

3 Method

3.1 Model Architecture

The proposed model is a **3D Residual Attention U-Net**, a volumetric deep learning architecture that extends the standard U-Net framework through two primary modifications: *Residual Learning* and *Attention Gating*.

- **Residual Encoder–Decoder:** Standard convolutional blocks are replaced with Residual Blocks in both the contracting and expanding paths. Each block consists of two $3 \times 3 \times 3$ convolutions, followed by Instance Normalization and LeakyReLU activation ($\alpha = 0.1$). The residual formulation improves gradient propagation, stabilizes deeper training, and enables the network to learn more expressive feature representations without degradation.

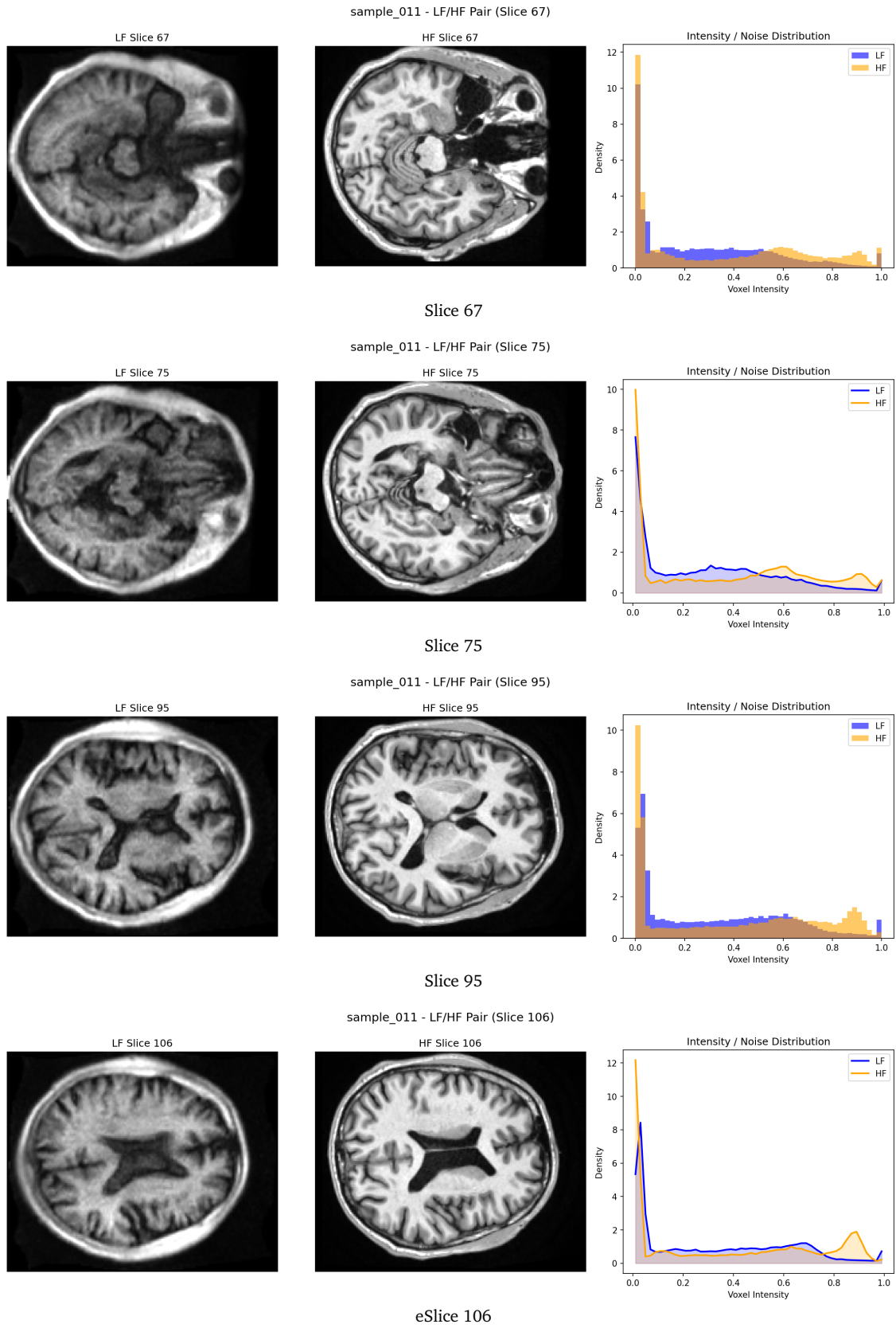


Figure 1: Full Slice Sequence Analysis for Sample 011. Observing the 64mT LF slices (left), the 3T HF slices (middle), and the Intensity Distribution of the LF and HF slices (right).

- **Attention Gating (AG):** At each skip connection, an Attention Gate filters encoder feature maps using the decoder’s gating signal. A sigmoid-based attention mask suppresses irrelevant spatial activations and emphasizes salient regions before feature concatenation. This mechanism encourages the model to focus on anatomically meaningful structures and reduces noise propagation through skip connections.

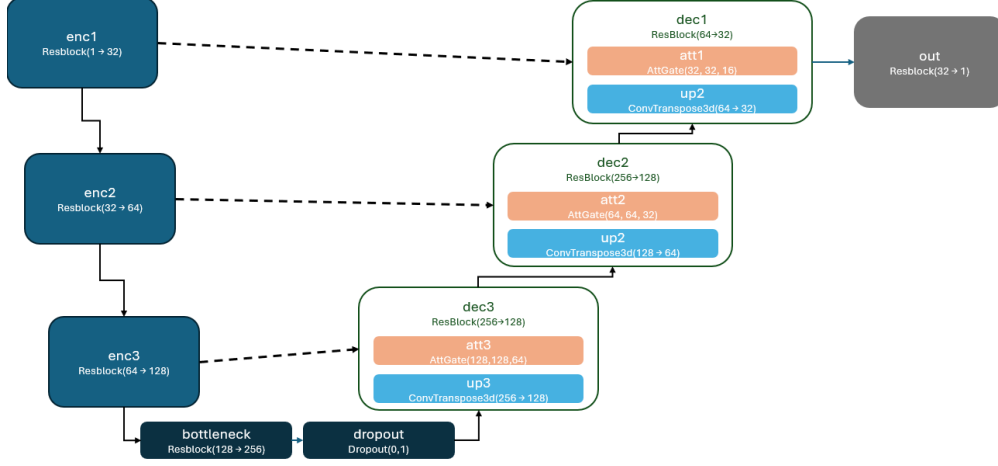


Figure 2: Diagram of project UNET3D model. Blue blocks are encoder down blocks, dark blue is the bottleneck, cyan is a convolutional transpose operations, orange are attention gates, green boxes (no fill) are decoder blocks, and the output is in the gray block. Solid lines represent connections while dotted lines are skip connections.

3.2 Training Strategy and Experimental Protocol

Throughout both the full 3D and the 2.5D modeling phases, we maintained a consistent experimental protocol to ensure fair comparison across architectural variants.

3.2.1 Data Split

All experiments were conducted using an 80/20 train-validation split at the subject level to prevent data leakage across slices belonging to the same volume. This ensured that validation performance reflected generalization to unseen volumetric data rather than memorization of adjacent slices.

3.2.2 Loss Function Exploration

Across both 3D and 2.5D architectures, multiple loss formulations were explored to balance pixel-wise fidelity and structural similarity. These included:

- $L1$ loss for robust pixel-level reconstruction,
- Mean Squared Error (MSE),
- Structural Similarity Index (SSIM),
- Hybrid combinations such as:

$$\mathcal{L} = \lambda_1 L1 + \lambda_2 (1 - SSIM),$$

- Metric-driven refinements incorporating PSNR as an auxiliary objective.

The motivation behind these experiments was to align the optimization objective with the competition evaluation metric while preserving perceptual and structural fidelity. Despite architectural transitions from 3D to 2.5D, the loss exploration remained consistent, allowing us to isolate the impact of spatial modeling choices independently from objective-function variations.

3.2.3 Incorporation of MS-SSIM Under Metric Evolution

During the course of the competition, the evaluation metric underwent changes that increasingly emphasized structural similarity rather than purely pixel-wise fidelity. In response to this shift, we incorporated Multi-Scale Structural Similarity (MS-SSIM) into our training objective.

Unlike single-scale SSIM, MS-SSIM evaluates structural similarity across multiple resolutions, making it more sensitive to perceptual quality and multi-scale anatomical consistency. This was particularly relevant for volumetric reconstruction tasks where structures appear at varying spatial scales.

We therefore modified the loss function to include a weighted MS-SSIM component:

$$\mathcal{L} = \lambda_1 L1 + \lambda_2 (1 - \text{MS-SSIM}),$$

where λ_1 and λ_2 were tuned empirically.

This adjustment allowed the optimization process to better reflect the competition’s scoring mechanism. Notably, the incorporation of MS-SSIM yielded measurable improvements in validation structural similarity, particularly in the 2.5D setting where slice-level evaluation was directly aligned with the metric formulation. The integration of MS-SSIM highlights the importance of adapting training objectives to evolving evaluation criteria, ensuring that optimization remains metric-aware rather than solely reconstruction-driven.

3.3 Additional Optimization Attempts and Future Directions

Given the limited improvements obtained from incremental refinements of the 3D architecture, we shifted our focus toward alternative modeling paradigms, including a Generator-Refiner pipeline, Generative Adversarial Networks (GANs), 2.5D U-Net architectures, and an attempt at Diffusion based models. These approaches aim to balance volumetric context modeling with improved training stability and generalization.

3.3.1 Architectural Shift: From Full 3D to 2.5D Modeling

Initial Full 3D Approach

The project initially adopted a fully 3D Residual Attention U-Net architecture operating on volumetric inputs using `nn.Conv3d`. This design explicitly modeled spatial dependencies across all three axes, allowing the network to capture rich inter-slice correlations and volumetric structural continuity.

While this approach was theoretically well-suited for volumetric reconstruction, it introduced substantial computational overhead and increased memory constraints. More importantly, the competition evaluation metric was computed on a slice-wise basis rather than volumetrically.

Motivation for Transition to 2.5D

Although the full 3D model captured deep volumetric context, the evaluation protocol primarily rewarded per-slice reconstruction quality. As a result, certain volumetric features learned by the 3D model were not directly reflected in the competition score.

To better align the model’s inductive bias with the evaluation metric, we transitioned to a 2.5D framework. Instead of processing entire volumes simultaneously, the model operates on stacked adjacent slices, using limited depth context while still predicting slice-level outputs.

This strategy preserves local inter-slice information while optimizing specifically for slice-wise reconstruction fidelity.

Architectural and Practical Advantages

The shift to 2.5D offered some theoretical advantages:

- Improved alignment between training objectives and evaluation metrics.
- Faster training iterations and easier experimentation.
- Some observed enhanced sensitivity to slice-level structural details.

Resulting Design Philosophy:

The final 2.5D architecture represents a metric-aware modeling decision rather than a simplification. By constraining volumetric context to localized depth neighborhoods, the model balances contextual awareness with evaluation alignment. However, these design improvement expectations did not materialize in score improvements.

3.3.2 Generator-Refiner Strategy

Given the strong baseline of the 3D UNET, we decided a good potential approach was to train another model that would act as a voxel "refiner" to improve the structural accuracy of the model and provide more fine-detail.

Design Idea: The refiner would utilize a pre-trained model (attempted using SMP UNET and Real-ESRGAN) and focus heavily on improving the model's PSNR score in order to improve voxel accuracy.

Shortcomings: Although good in theory the models usually did not create any meaningful improvements in the seen PSNR score and continued to produce "safe" blurry voxels that "best fit" the volumes without taking any meaningful risks.

3.4 Attempt at Diffusion Model:

Attempting to copy a more state-of-the-art model we heavily referenced the work seen in the article "Taming diffusion transformers for high-fidelity MRI super-resolution" [9] in which they proposed the "Diffusion Transformer for MR image super-resolution" (DiTMSR). Although the idea was positive, lack of experience and likely poor execution of the pre-processing and pipeline led to this branch not being explorable due to time constraints. The model Tu and his team proposed is state-of-the-art but we were unable to replicate it in the time frame.

3.4.1 Other Attempted Refinements:

Given the strong baseline performance of the 3D Residual Attention U-Net, we explored several strategies to further improve reconstruction quality. However, most of these modifications did not yield measurable improvements.

1. **Two-Stage Fine-Tuning Strategy:** We implemented a two-step training procedure. In Stage 1, the model was optimized using the combined loss:

$$\mathcal{L} = L1 - (1 - SSIM).$$

In Stage 2, early layers were frozen and additional MSE and PSNR objectives were introduced to explicitly improve these metrics. Although training remained stable, this approach resulted in degraded validation performance, suggesting metric interference and potential over-regularization.

2. **Learning Rate Scheduling and Early Stopping:** A learning rate scheduler and early stopping were introduced based on validation SSIM and the competition evaluation metric. While training became more controlled, no consistent improvement in final performance was observed.
3. **Data Augmentation:** Spatial augmentations were applied to increase data diversity. This produced slight improvements but did not significantly alter model performance.
4. **EMA, K-Fold Validation, and Ensembling:** Exponential Moving Average (EMA), cross-validation, and model ensembling were incorporated to improve generalization. Despite the increased computational cost, these techniques did not produce meaningful gains in the evaluation metric.

4 Results

4.1 Training Details

The models were trained on a high-performance compute cluster using SLURM (slurm_job.sh, slurm_job_metric.sh).

4.1.1 Hyperparameter Tuning Observations

- **Adam Learning Rate:** Smaller rates (1×10^{-5}) led to excessively slow convergence, while larger rates (1×10^{-3}) caused the loss to oscillate and occasionally diverge. We tested many intermediate learning rates between these two ranges but The final rate of 2×10^{-4} which provided the best stability and final score.
- **Batch Size:** Due to the memory demands of 3D convolutions, testing batch sizes larger than 16 (e.g., 32 or 64) resulted in Out-Of-Memory (OOM) errors on some of our training GPUs unless patch sizes were reduced. Within testing batch sizes of 4, 8, and 16, the batch size of 4 offered the best compromise between gradient stability and memory utilization.
- **Volume Patch Size (ROI Size):** Smaller patches (32^3) prevented the network from learning broader spatial context and structural dependencies in the MRI scans. Conversely, larger patches (128^3) severely limited the allowable batch size and samples per volume. Our two most competent patch sizes were that of 64^3 and 96^3 , through testing both we saw that the 96^3 was optimal for capturing sufficient anatomy.
- **Model Base/Expressiveness** The base value in our 3D UNET determined how many "features" the model the model can learn at each level. Here we tested many different options ranging from 32 to 96, with the optimal feature was found at 56. We believed this was a good intermediary value that would allow the model to capture enough complexity and have enough feature detection without getting overly complicated. Expanding this number with the lack of volume data led to underperforming models that overfit specific attributes which could be easily visualized during validation loops.
- **Number of Epochs:** Due to the low amount of data, training for more than 40-50 epochs left the model overfit. Due to this we mostly stuck to low epoch count training runs and attempted to get more generalizable models.
- **Scheduler:** We tested both a Cosine Annealing LR as well as a Reduced LR on Plateau. Both obtained similar results but the Reduced LR on Plateau gave us a little more control on the LR decay and ended up with a slightly better performance in the submission.
- **Frameworks:** PyTorch.
- **GPUs:** 1 A100 GPU.
- **Hyperparameters:** The generator and discriminator were optimized using the Adam optimizer with a learning rate of $2e - 4$. The batch size was set to 4 3D patches per step with 96 voxels per patch.
- **Loss Function:** The total loss function was a weighted combination of:
 - Pixel Loss (L1/L2) to ensure overall structural accuracy.
 - Perceptual Loss derived from our `metric_model`.
- **Scheduler:** Reduced LR on Plateau with a learning rate decrease factor of 0.5, a patience of 5 epochs and a minimum LR of $1e - 5$.
- **Validation:** Validation was performed every epochs (`slurm_job_eval.sh`), tracking structural similarity and peak signal-to-noise ratio.

4.2 Results

The performance of our proposed model was evaluated against traditional baseline methods (e.g., Trilinear Interpolation). We utilized standard quantitative metrics adapted for 3D data:

In our validation dataset:

- **Peak Signal-to-Noise Ratio (PSNR):** $16.02 \pm 0.27\text{db}$

- **Structural Similarity Index Measure (SSIM):** $72.8\% \pm 0.4\%$
- **Combined Kaggle 2 Submission Score** 0.5992

Below some of the created output images can be visualized and compared to both the original LF slice and the "ground truth" HF slice.

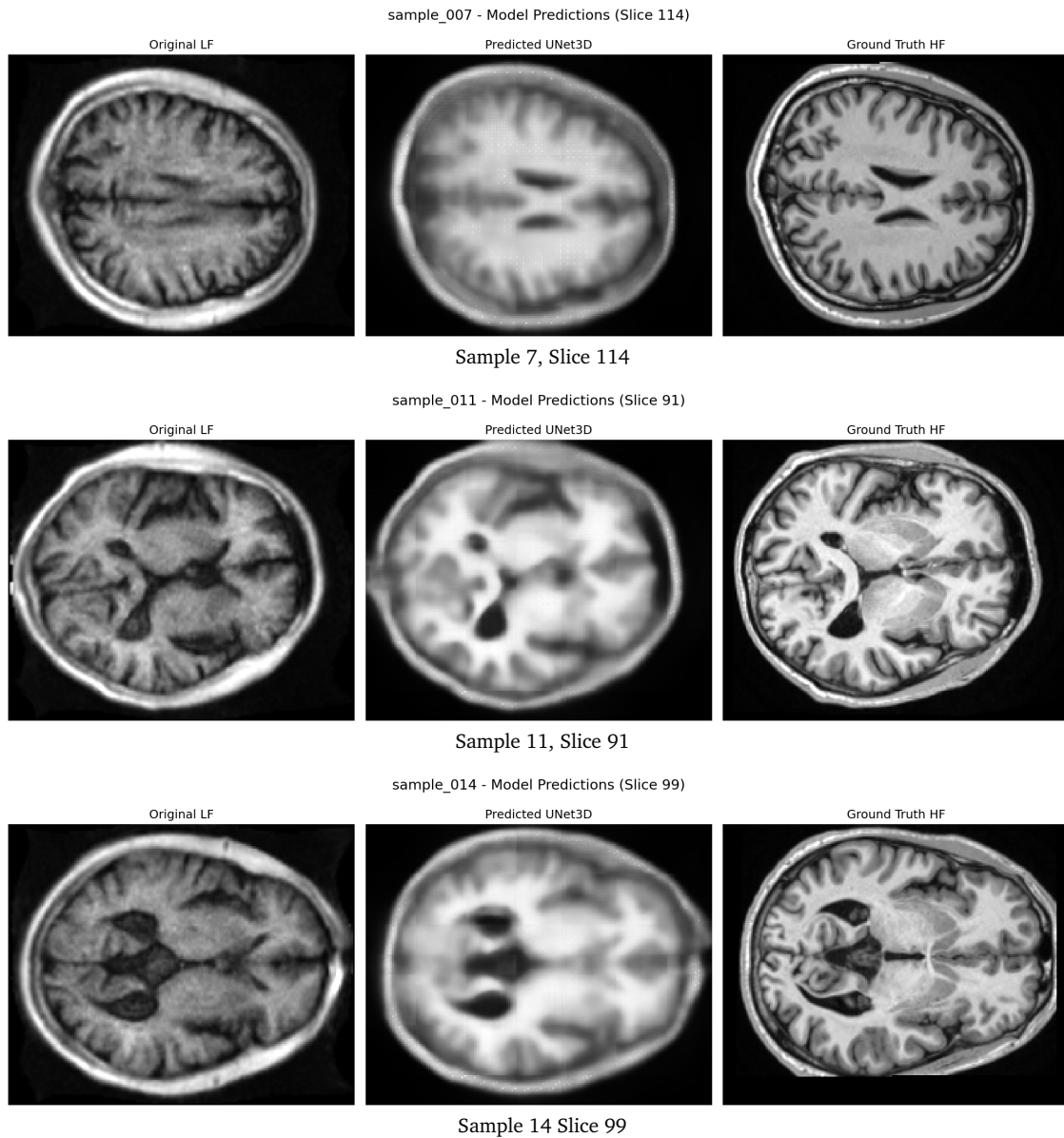


Figure 3: Slice Output Analysis for multiple samples. Observing the 64mT LF slices (left), the generated model slice (middle), and the ground truth 3T HF slices (right).

Observing the above images in 3 we can observe that in some instances the model began to see some of the complexities in the higher end MRI scans but failed to remove its "blurriness".

Looking at this from another angle, exploring the differences directly, we can plot a heatmap of differences in slices to observe:

In figure 4 we can more clearly observe the fact that the model did not have any massive errors in its predictions, given the blurry nature of its safe guesses. However, we can also observe that the model did "learn" certain aspects of the brain that it corrected for. Looking at our sample 14 image, where the brain MRI cut off the bottom part of the skull and likely had some sort of obstruction during the imaging, the model seemingly correctly filled in the empty space with a continuous anatomical structure that does not look out of place.

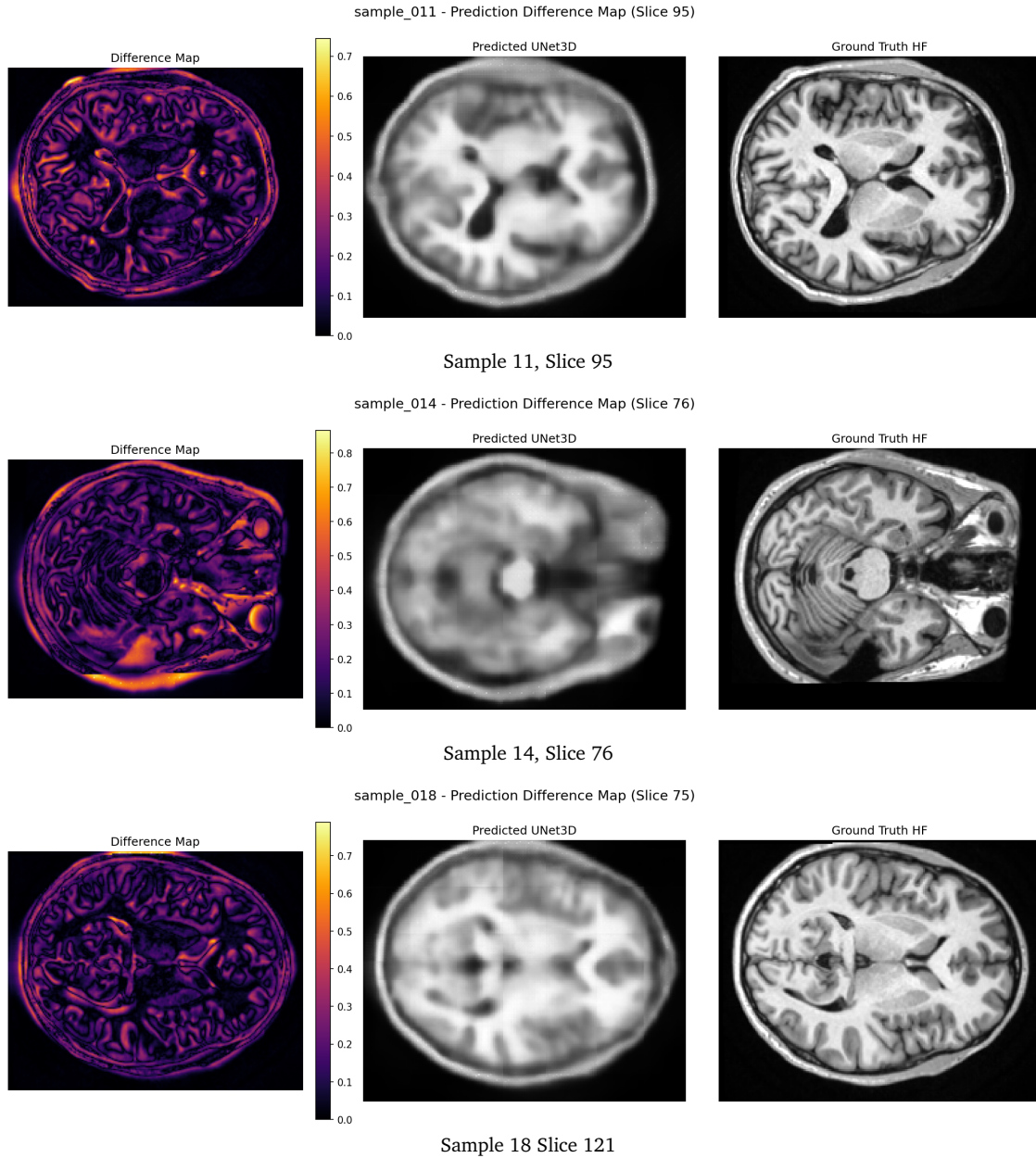


Figure 4: Slice Output Analysis for multiple samples. Observing the difference heatmap of the slices (left), the generated model slice (middle), and the ground truth 3T HF slices (right).

5 LLM Prompts

- Idea generation of which models could be useful (2.5D UNET, 3D UNET, variety of GAN models, diffusion attempt) based on article blah blah.
- Zero shot creation of visualization prompts.
- Conversations regarding performance and loss evaluation. After creation of original model and the other attempts we had conversations with LLMs so that we could quickly gather hyperparameter tuning ideas, diverse loss functions, and other changes we could make to try and improve performance.

6 Conclusions

Our 3D U-Net architecture, along with subsequent experimental iterations, underperformed relative to established benchmarks by a significant margin—specifically exceeding 100bps in SSIM and 20dB in PSNR.

While we achieved a moderate SSIM score of 72.8%, suggesting the model effectively captured coarse structural information from high-resolution reference images, it struggled with high-frequency details. The model tended to prioritize "safe" predictions, resulting in blurred voxels and extreme intensity values that optimized the scoring scheme but lacked anatomical precision. This performance plateau across multiple architectures strongly suggests that our primary bottleneck was an under-explored preprocessing pipeline. The convergence of different models toward similar scores was a clear indicator that the limiting factor was the data representation rather than the model capacity.

Future Work:

Future iterations should pivot focus from architectural tuning to extensive data preprocessing and augmentation. Our project demonstrated that regardless of the model complexity or loss function applied, the outputs consistently suffered from "blurring" artifacts—maintaining an SSIM in the mid-70s while failing to improve a stagnant PSNR (15dB).

- To move beyond these "safe" but low-fidelity predictions, upcoming work should prioritize:
- Advanced Normalization: Investigating more robust voxel intensity scaling and noise reduction.
- Feature Engineering: Better alignment and registration of input volumes to reduce spatial ambiguity.
- Loss Function Evolution: While previous attempts to modify loss functions were insufficient, combining improved data quality with perceptual or GAN-based losses may better incentivize the recovery of fine details.

Ultimately, the results underscore a fundamental machine learning principle: architectural complexity cannot compensate for a sub-optimal data pipeline.

7 Author contributions

- Overall, both authors contributed across the project with the only difference being Thiago Viegas creating most of the preprocessing framework. Other than that both teammates contributed equally to training a variety of different models, hyperparameter tuning, and experimentation in general.

8 References

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